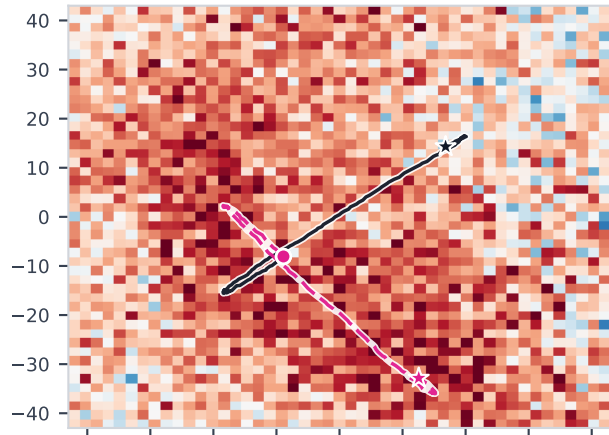


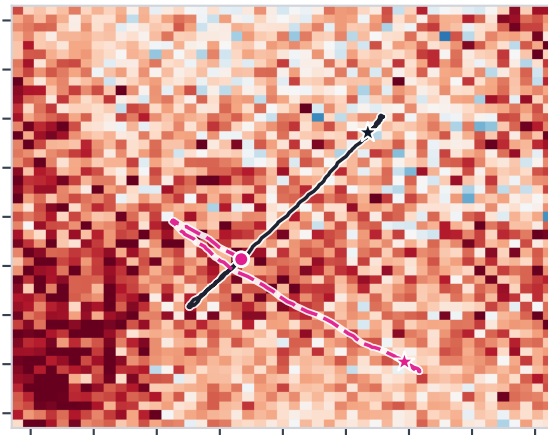
What learning adds: lift = $f(\theta)$ |RL on – $f(\theta)$ |RL off, same θ , same 5 maps — baseline environment

Evolution — weakest runs
($f < 0.272$, a median split
— still above viability)

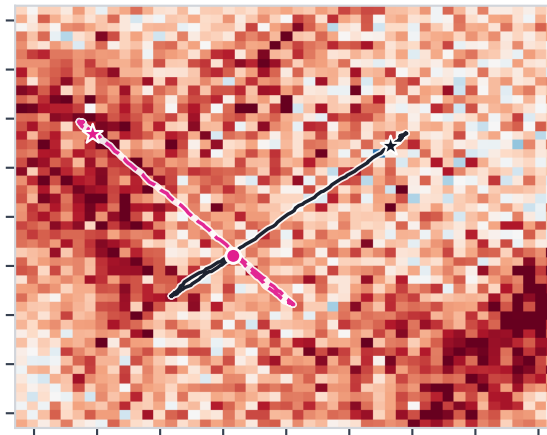
seed 2007 · run $f = 0.268$ · plane 64%
median lift +0.023



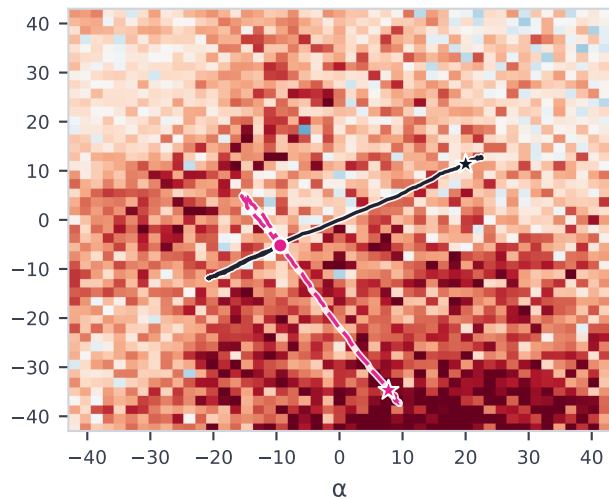
seed 2004 · run $f = 0.267$ · plane 64%
median lift +0.023



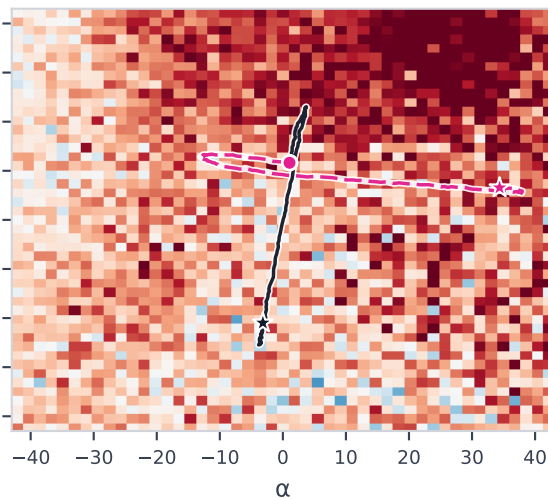
seed 2003 · run $f = 0.265$ · plane 64%
median lift +0.024



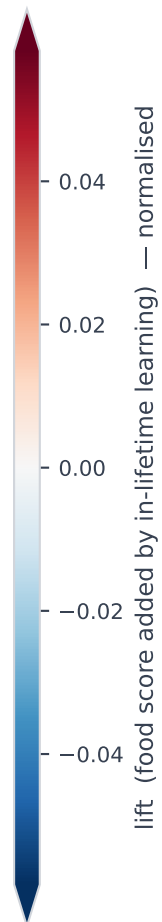
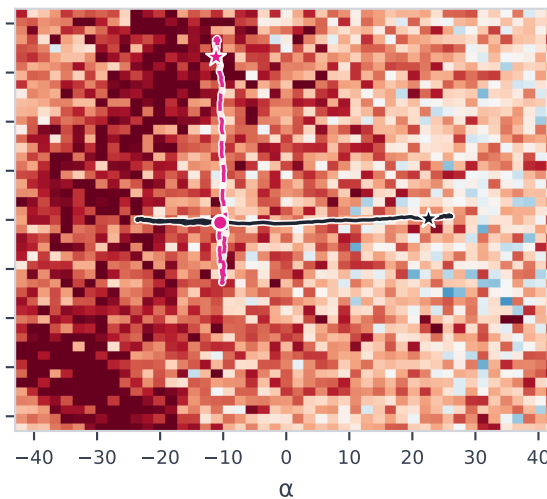
seed 2002 · run $f = 0.277$ · plane 63%
median lift +0.023



seed 2006 · run $f = 0.276$ · plane 63%
median lift +0.025



seed 2008 · run $f = 0.275$ · plane 63%
median lift +0.029



Evolution — succeeded

— evolution run - - - learning run ● gen 0 ★ final generation